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MECHANIC GUIDE

Aquazone UV equipments

Aquaz-S6

Aquaz-S25

Aquaz-S55

Aquaz-S165

Aquaz-S165-B

Aquaz-S275-B

Aquaz-S660-B

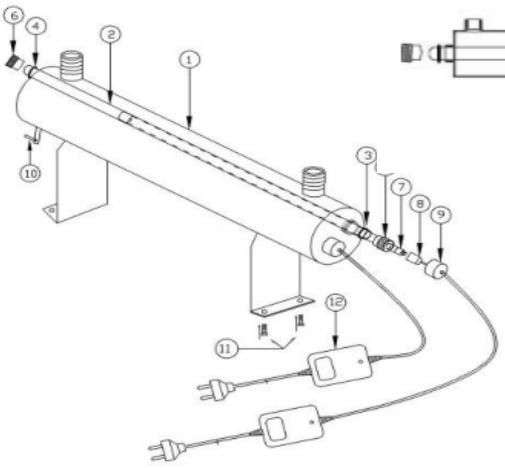
Befure using the equipment, please read the guide carefully!

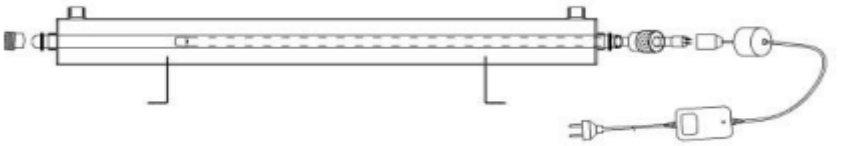
TABLE OF CONTENTS

1. UV equipments
2. UV equipment's main appurtenances
3. Required UV dose for inactivity
4. Assembly
5. Operation
6. Important safety regulations
7. The system's maintenance
8. Typical wiring illustration
9. Electronic wiring illustration

1. UV equipments

Description:	Stainless steel UV sterilizer, for water treatment.
Application:	Drinking water, food production and handling, health care, industry, etc.
Usage:	Stops the reproduction of bacteria, viruses, mushrooms and algae in water.
Types:	Aquaz-S6, Aquaz-S25, Aquaz-S55, Aqua165S,Aquaz-S165-B, Aquaz -S275-B, Aquaz-S660-B;
Wattage:	6-660W
Power supply:	230V 50/60Hz





1 Reactor
2 Quartz sleeve
3 Silicone sealing
4 Silicone sealing
5 Nut - open
6 Nut - closed
7 UV lamp
8 4-pin electrical connector to UV lamp
9 End cap
10 Drain tap (from Aquaz-S165 type)
11 Mounting parts
12 Power supply

Part number	Products	Conn.	UV Lamp pcs x W	Sizes H x W x L (mm)	Flow rate 16 mJ/cm² (m³/h)	Flow rate 30 mJ/cm² (m³/h)	Flow rate 40 mJ/cm² (m³/h)
Aquaz-S6	UV equipment, household sterilizing	1/4"	1 x 6W	63 x 50.8 x 270	0.4	0.2	0.2
Aquaz-S25	UV equipment, household sterilizing	1/2"	1 x 25W	82 x 63.5 x 595	3.1	1.7	1.3
Aquaz-S55	UV equipment, household sterilizing	3/4"	1 x 55W	85 x 63.5 x 950	6.9	3.7	2.8
Aquaz-S165	UV equipment, household sterilizing	1,5"	3 x 55W	283 x 133x 950	20.6	11	8.3
Aquaz-S165-B	UV equipment, household sterilizing with control box	1,5"	3 x 55W	283 x 133x 950 400 x 300 x 150	20.6	11	8.3
Aquaz-S275-B	UV equipment, household sterilizing with control box	2"	5 x 55W	283 x 133 x 950 400 x 400 x 150	34.4	18.3	13.8
Aquaz-S660-B	UV equipment, household sterilizing with control box	3"	12 x 55W	430 x 220 x 950 500 x 500 x 250	82.5	44	33

2. UV equipment's main appurtenances

Part number	Products
Aquaz-AD-S	Adapter 6W, 25W, 55W, 165W, 275W, 660W UV system
Aquaz-OR-S	O-ring 6W, 25W, 55W, 165W, 275W, 660W UV system

Part number	Products
QS-6W	Quartz sleeve S6 system.
QS-25W	Quartz sleeve S25 system
QS-55W	Quartz sleeve S55-S660 system

Part number	Products
S6WRL	UV lamp S6
S25WRL	UV lamp S25
S55WRL	UV lamp S55-S660

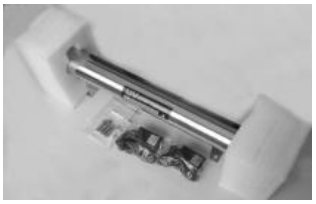
3. Required UV dose for inactivity

UV Dose required for Inactivation	
Patogen	**Efficiency 4-Log (99.99%)**
Giardia lamblia cysts	1.7 mJ/cm^2
Vibrio chloerae	2.9 mJ/cm^2
Shigella sysenteriae	3.0 mJ/cm^2
Eschericha coli 0 157:H7	5.6 mJ/cm^2
Cryptosporidium parvum cocvsts	5.7 mJ/cm^2
Salmonella typhi	8.2 mJ/cm^2
Shigella sonnei	8.2 mJ/cm^2
Legionella pneumophila	9.3 mJ/cm^2
Salmonella enteritidis	10.0 mJ/cm^2
Hepatitis A virus	29.6 mJ/cm^2
Poliovirus Type 1	30.0 mJ/cm^2
Coxsackie B5 virus	30.0 mJ/cm^2
Rotavirus SA 11	36.0 mJ/cm^2
Adenovirus	186.0 mJ/cm^2

	UV Dose @ 95% UVT			UV Dose @ 85% UVT		
m³/h	**16 mJ/cm²**	**30 mJ/cm²**	**40 mJ/cm²**	**16 mJ/cm²**	**30 mJ/cm²**	**40 mJ/cm²**
0	Aquaz-S6	Aquaz-S6	AquazS6	Aquaz-S6	Aquaz-S6	Aquaz-S6
0.24					S5Q	S5Q, S8Q
0.48		S5Q	S5Q	S5Q		
0.68	Aquaz-S25, S5Q					S8Q, Aquaz-S25
0.91					S8Q, Aquaz-S25	Aquaz-S25
1.14						
1.37			Aquaz-S25, S8Q	S8Q, Aquaz-S25		Aquaz-S55
1.82		Aquaz-S25, S8Q			Aquaz-S55	Aquaz-S55
2.28			Aquaz-S55			VH410/2
2.73		Aquaz-S55	Aquaz-S55 VH410/2		Aquaz-S55 VH410/2	VH410/2
3.18	Aquaz-S25, S8Q			S8Q		Aquaz-S165
3.64	S8Q	Aquaz-S55 VH410/2	Aquaz-S165	AquazS55	VH410/2	
4.09					Aquaz-S165	
4.55	Aquaz-S55	Aquaz-S165				
5				Aquaz-S55 VH410/2		
5.45						
5.91	Aquaz-S55 VH410/2			VH410/2		
6.36				Aquaz-S165		
6.82						Aquaz-S275
7.27						
7.73	Aquaz-S165					
8.63			Aquaz-S275		Aquaz-S275	
9.54						
10.5						Aquaz-S660
11.4		Aquaz-S275				
13.6						
15.9			Aquaz-S660		Aquaz-S660	
20.5		Aquaz-S660		Aquaz-S275		
24.9	Aquaz-S275					
29.5				Aquaz-S660		
34.1						
38.6	Aquaz-S660					
43.2						
54.5						
72.7						

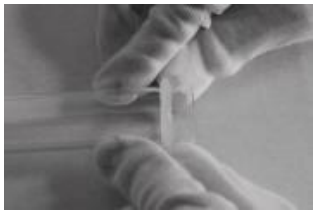
	UV Dose @ 75% UVT			UV Dose @ 50% UVT		
m³/h	**16 mJ/cm²**	**30 mJ/cm²**	**40 mJ/cm²**	**16 mJ/cm²**	**30 mJ/cm²**	**40 mJ/cm²**
0	Aquaz-S6	S5Q, Aquaz-S25	S5Q, Aquaz-S25	S5Q	S5Q	S8Q, Aquaz-S25
0,24					S5Q , Aquaz-S25	
0,48	S5Q , Aquaz-S25			S5Q , Aquaz-S25	S8Q, Aquaz-S25	S8Q
0,68			S8Q, Aquaz-S25		S8Q	VH410/2, Aquaz-S55
0,91		S8Q, Aquaz-S25	S8Q Aquaz-S55	S8Q, Aquaz-S25	VH410/2, Aquaz-S55	
1,14		S8Q Aquaz-S55		S8Q , Aquaz-S55		VH410/2
1,37						
1,82	S8Q Aquaz-S55		VH410/2	VH410/2, Aquaz-S55	VH410/2	Aquaz-S165
2,28		VH410/2	Aquaz-S165	VH410/2	Aquaz-S165	
2,73						
3,18	Aquaz-S55	Aquaz-S165				Aquaz-S275
3,64				Aquaz-S165		
4,09	VH410/2				Aquaz-S275	
4,55						Aquaz-S660
5	Aquaz-S165					
5,45			Aquaz-S275			
5,91					Aquaz-S660	
6,36		Aquaz-S275		Aquaz-S275		
6,82						
7,27						
7,73						
8,63			Aquaz-S660			
9,54						
10,5		Aquaz-S660		Aquaz-S660		
11,4	Aquaz-S275					
13,6						
15,9						
20,5	Aquaz-S660					
24,9						
29,5						
34,1						
38,6						
43,2						
54,5						
72,7						

4. Assembly

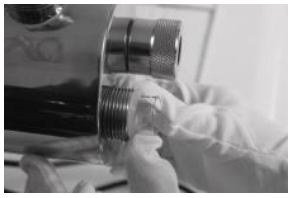


1. Open and check the content of the packet: reactor, UV lamp, quartz sleeve, silicone sealing and power supply.
2. Before assembling check that the UV lamp and the quartz sleeve is clean and free of fingerprints. Remove the dirt with gentle alcohol.

3: Screw off the open and closed nut from the reactor

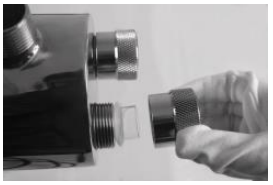


4: Put the silicone sealing on the open end of the quartz sleeve. Ideal distance is 12mm from the end of the sleeve. Depending on the type the sealing can be O-ring or split conical



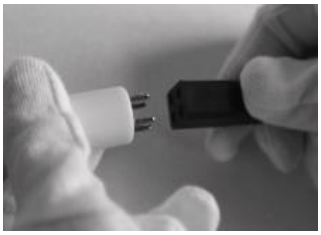
5: Put the quartz sleeve into the reactor with the closed end ahead

6: Put the silicone sealing on the other end of the quartz sleeve. Slide the conical sealing into the groove on both end of the reactor. Take care: the closed end of the quartz sleeve should not extend from the reactor more than12mm. When screwing the nut forcing can cause the quartz sleeve to crack.



7: Carefully screw the closed nut.

8: Carefully screw the open nut. Take care not to crack the quartz sleeve.



9: Connect the 4-pin connector to the UV lamp tightly.

10: Carefully put the UV lamp into the quartz sleeve through the open nut.

11: Put the end cap on the open nut.

12: In case of two or more UV lamp type systems repeat the above procedure.

13. After mounting and connecting to the water system carefully open the tap and check whether there is leakage or not.
14. Choose an appropriate place for the disinfection system on the wall or on a solid flat surface. It is recommended to mount the UV system vertically because of the better ventilation. When mounting take into consideration the place requirement of the UV lamp replacement.
15. Let the water flow for a couple of minutes to deaerate the system and to remove dirt.
16. Connect the power supply to the power source.

5. Operation

Water quality

The most important operational requirement of the UV disinfection system is the optically clean, turbidity-free, colourless water.

- Water temperature: 2-45^{o}C
- Fe content: <0.3ppm (0,3mg/l)
- Water hardness: <6,73°dH
- Turbidity: <1NTU
- Mn content: <0,05ppm (0,05mg/l)
- UV transmittance: > 75%
- Befure turning on, please check the power plug.

a) Do not use the UV disinfection system with damaged UV lamp or quartz sleeve.
b) The UV disinfection system is planned for continuous operation. Repeated ON-Off switching considerably decreases the power (effectiveness) of UV radiation and shortens the service life of the equipment.
c) Immediately turn off the main switch of the UV disinfection system in case of soaking and do not use it again.
d) Do not operate the equipment if its power plug or cord is damaged, if it operates improperly of is damaged.
e) A filter shall be installed upstream the UV.
f) Under 2 °C temperature disconnect from the power source and drain the water from the reactor.

6. Important safety regulations

	Comply with the electrical regulations.
	The UV radiation is not visible for the naked eyes but it is harmful for the skin and eyes. Never look directly into the UV light during the operation of the equipment. Do not pull out the UV lamp before switching off the equipment. If it is unavoidable use safety glasses.
	Keep the basic safety rules in order to prevent fire and electrical shock. Makes sure that electric shock protection regulations are kept.
	Do not touch the UV lamp during operation or directly after operation in order to avoid burns. Use safety gloves.

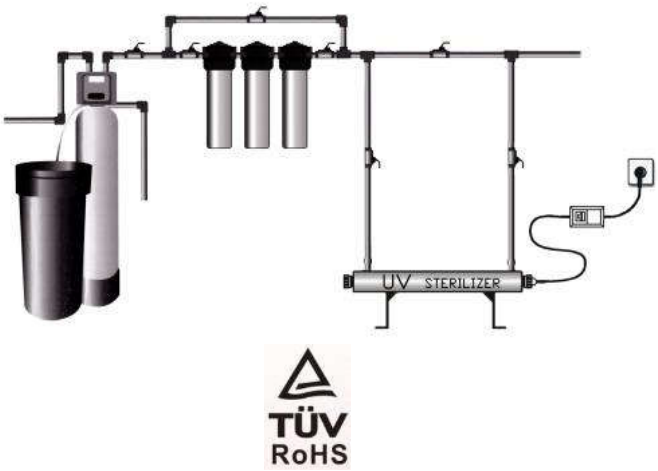
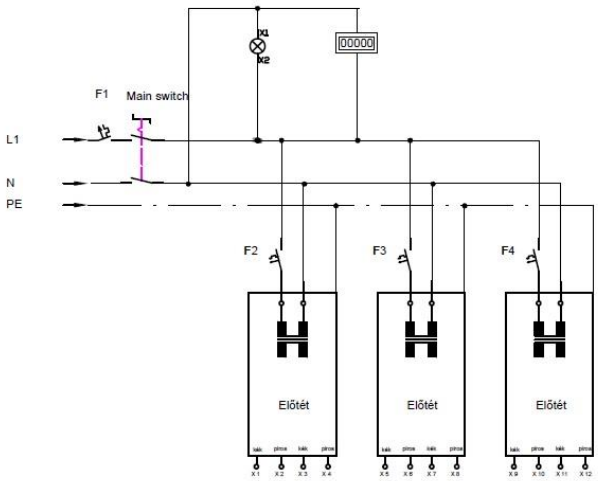
7. The system's maintenance

a) At least monthly ones check the signal light of the power supply. The red LED light indicates the proper operation. The green LED indicates the proper power connection. When the red LED light turns off replace the UV lamp immediately. When the green light turns off immediately replace the power supply.

b) It is recommended to replace the UV lamp after 8000 working hour or after one year. After 8000 working hour the UV lamp may light but its performance might be significantly lower.

c) Depending on the contamination of the water clean the quartz sleeve after 3-6 month of working. Use alcohol based detergents and gentle sponge.

8. Typical wiring illustration



9. Electronic wiring illustration